

COMEX GOLD FUTURES (GC)

Spread & Curve Structure — Fundamental Reference

Real Rates, Central Banks, USD, Geopolitics, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of COMEX Gold futures (ticker: GC), contract size 100 troy ounces, with particular focus on the forces that shape the forward curve and the spread structure between contract months. Gold occupies a unique position in the commodity universe: it is simultaneously a mined physical commodity, a monetary asset with thousands of years of history, a central bank reserve asset held by virtually every sovereign state, an industrial material, and a financial instrument whose price responds more to macro-financial variables than to traditional supply and demand fundamentals. Unlike petroleum or agricultural markets where supply shocks and demand cycles dominate, gold pricing is primarily determined by the opportunity cost of holding a non-yielding asset — real interest rates — and by its role as a store of value and safe haven in periods of financial stress, currency debasement, geopolitical risk, and monetary policy uncertainty. Understanding this macro-financial framework, alongside the physical gold market mechanics, the carry structure of the futures curve, and the cross-market relationships that anchor gold prices, is the foundation of spread and butterfly trading in this market.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and the Physical Gold Market

COMEX Gold futures are the world's primary exchange for gold price discovery and risk transfer. The contract is physically deliverable, creating a direct link to the global OTC gold market centred in London (the LBMA).

Parameter	Detail
Exchange	CME Group — COMEX division (New York)
Ticker	GC (front month); also traded as Gold on ICE and via LME forwards
Contract size	100 troy ounces
Quotation	USD per troy ounce; minimum tick = \$0.10/oz = \$10.00 per contract
Contract months	February (G), April (J), June (M), August (Q), October (V), December (Z) — six even-month contracts per year. Also active: nearby odd months (H, K, N, U, X) for the nearest 3 months. Active contracts extend approximately 3 years forward; liquid concentration in front 6–8 contracts.
Settlement type	Physical delivery of 100 troy oz gold bars (minimum 99.5% fineness, COMEX-approved assayers) at COMEX-approved depositories in New York/New Jersey or Delaware. Approximately 95%+ of contracts are cash-settled or rolled; physical delivery occurs but is a small fraction of open interest.
Trading hours	CME Globex: Sunday–Friday 18:00–17:00 ET (23-hour session); liquid hours approximately 08:00–13:30 ET (overlap with LBMA London session)
Last trading day	Third-to-last business day of the delivery month
Contract value at \$2000/oz	\$200,000 per contract
DV01 equivalent	Not applicable in conventional sense; sensitivity to 1bp change in real rates $\approx$ variable (see Section 3)
LBMA relationship	London Bullion Market Association (LBMA) runs the global OTC gold market. Twice-daily London Gold Price Fix (AM and PM) is the primary physical gold benchmark. COMEX GC futures and LBMA spot price are closely arbitrated; the COMEX-LBMA spread (EFP — Exchange of Futures for Physical) is a key structural variable in the spread market.

1.1 The LBMA and the Global OTC Gold Market

- **London as the global gold trading hub:** the LBMA unallocated gold market in London is the world's largest gold trading venue by volume. London daily clearing volumes typically exceed 20–25 million troy ounces (600–800 tonnes), far exceeding COMEX futures volumes on a gold-equivalent basis. London spot prices are the primary reference for physical gold transactions globally.
- **The London Gold Price (LBMA Gold Price):** administered by ICE Benchmark Administration (IBA). Two fixings per day: AM (10:30 London time) and PM (15:00 London time). Electronically auctioned process replacing the historical "Gold Fix" telephone conference. The PM fixing is the most widely used benchmark for physical transactions, ETF NAV calculations, mine hedging, and central bank valuations.
- **The EFP (Exchange of Futures for Physical):** the spread between COMEX GC futures price and the LBMA spot price. In normal conditions the EFP is small and positive (COMEX at a slight premium to LBMA, reflecting carry to futures delivery). During stress events (COVID March 2020, Brexit-related flows), the EFP widened dramatically as physical demand in London diverged from paper gold pricing in New York. The EFP is the primary signal of dislocation between physical and futures gold markets.

1.2 Allocated vs Unallocated Gold

- **Unallocated gold:** a claim on a bullion bank for a quantity of gold without specific bars being designated. Most LBMA trading is in unallocated form. Unallocated accounts bear counterparty risk (if the bullion bank fails, the holder is an unsecured creditor).
- **Allocated gold:** specific gold bars held in custody, identified by serial number. No counterparty risk. COMEX physical delivery is in allocated bars held at approved depositories.
- **Gold lease rates and the forward rate:** gold can be lent (leased) by central banks to bullion banks, who pay a "gold lease rate" (GLR). The forward gold price is determined by: $\text{Spot price} \times (1 + \text{USD interest rate} - \text{Gold lease rate}) \times \text{time}$. When lease rates are low (gold is cheap to borrow), the forward curve is in contango determined primarily by USD rates. When lease rates spike (gold is scarce to borrow), the forward curve can flatten or go into backwardation.

2. Gold as a Monetary Asset — The Analytical Framework

Gold's price behaviour is fundamentally different from other commodities because it is primarily a monetary asset rather than a consumed commodity. The global above-ground gold stock (approximately 210,000 tonnes, accumulated over millennia) is approximately 60–70 times annual mine production. This means that unlike oil or wheat, annual production is a small fraction of the total stock — so supply shocks from mining are minor price drivers. Instead, the price is determined by shifts in the demand to hold existing gold stocks, which is primarily a function of macro-financial variables.

2.1 The Real Interest Rate Framework

The most robustly documented and theoretically grounded framework for gold pricing is the real interest rate model:

- **Gold as a non-yielding asset:** gold pays no interest, dividend, or coupon. Holding gold has an opportunity cost equal to the real interest rate on an alternative risk-free investment (typically measured by US Treasury Inflation-Protected Securities, or TIPS, yields). When real rates are negative or very low, the opportunity cost of holding gold is minimal and gold becomes relatively attractive. When real rates are high and positive, holding gold is expensive in opportunity cost terms and gold becomes relatively less attractive.
- **The documented relationship:** the price of gold and the US 10-year real yield (10-year TIPS yield) have a well-documented strong negative correlation. Gold rallied from ~\$700/oz to ~\$1,900/oz from 2008–2011 as real rates fell sharply. Gold fell from \$1,900 to \$1,050 from 2012–2015 as real rates rose from the taper tantrum. Gold rallied from \$1,050 to \$2,075 (record at the time) in 2018–2020 as real rates fell to negative territory. Gold sold off in 2021–2022 as the Fed hiked and real rates rose sharply. The relationship is not perfect — other factors periodically dominate — but it is the primary quantitative framework for gold valuation.
- **The formula:** as a rough approximation, a 100bp decline in 10-year real yields historically corresponds to approximately \$150–250/oz upward pressure on gold prices (the exact sensitivity varies with market conditions).
- **Short-term vs long-term real rates:** gold is most sensitive to real rates at the 5–10 year horizon (reflecting the typical investment horizon for gold holders). Short-term real rates (2-year TIPS) matter for near-term opportunity cost but are less important than the long-term structural rate level.

2.2 The USD Relationship

- **Negative correlation with the US Dollar Index (DXY):** gold is priced in USD. A stronger dollar makes gold more expensive in non-USD currencies, reducing demand from non-US buyers. A weaker dollar has the opposite effect. This creates a documented negative correlation between DXY and gold prices.
- **The correlation is not symmetric:** the USD-gold relationship is tighter when USD moves are driven by US real rate differentials (monetary policy divergence). When USD strengthens due to safe-haven demand (same flight-to-safety that supports gold), the correlation breaks down — both USD and gold can rally simultaneously. March 2020 demonstrated this: initially both USD and gold fell in the liquidity crisis, then both rallied on safe-haven and stimulus expectations.
- **Purchasing power of non-USD currencies:** for investors holding non-USD currencies, gold's return is the USD gold return plus the USD appreciation against their currency. When EM currencies weaken sharply (Argentina, Turkey, India), local demand for gold as a currency substitute increases, supporting global physical demand.

2.3 Inflation Expectations and Gold

- **Gold as an inflation hedge:** gold is widely held as protection against inflation. The empirical evidence for this is strongest over very long time horizons (decades) but weaker in the short to medium run (1-5 years). In the 1970s inflationary period, gold rallied dramatically. In the 2021-2022 high inflation period, gold initially underperformed because the Fed's aggressive rate hikes raised real rates simultaneously with nominal inflation — demonstrating that gold is a hedge against **negative real rates and currency debasement**, not simply against nominal inflation.
- **5-year 5-year forward breakeven inflation:** the market-implied long-run inflation expectation (derived from TIPS pricing). This measure of long-run inflation expectations is more relevant to gold than current CPI, because gold is held as a multi-year store of value. Rising long-run inflation expectations that are not fully offset by higher nominal rates (i.e. real rates fall) are bullish for gold.

3. The Forward Curve — Carry, Leasing, and the Contango Structure

Unlike most commodity markets where the forward curve reflects storage costs and seasonal demand patterns, gold's forward curve is primarily determined by the cost of carry — the difference between the USD interest rate and the gold lease rate. This makes gold spreads fundamentally a money market and interest rate product.

3.1 The Gold Forward Rate Formula

- **The theoretical forward price:** $F(T) = S \times \exp((r - l) \times T)$, where S = spot price, r = USD risk-free rate, l = gold lease rate (GOFO / SOFR-based), T = time in years. In practice, traders use the simplified: $\text{Forward} = \text{Spot} \times (1 + (\text{USD rate} - \text{lease rate}) \times \text{days}/360)$.
- **The GOFO (Gold Forward Offered Rate):** the rate at which bullion banks lend gold against USD (historically quoted by LBMA). GOFO was discontinued as a published rate in 2015; the equivalent concept is now expressed through the OTC gold swap market where the cost of swapping USD for gold (or vice versa) determines the effective gold forward rate. SOFR (secured overnight rate) has replaced LIBOR as the USD rate component.
- **Normal contango:** when USD rates exceed the lease rate (the normal condition), gold is in contango — the forward price exceeds the spot price. The magnitude of the contango equals approximately $(\text{USD rate} - \text{lease rate}) \times \text{spot price} \times \text{time}$. At \$2000/oz spot and USD rates of 5% with lease rates of 0.5%, the annual contango is approximately $4.5\% \times \$2000 = \$90/\text{oz}$ per year, or approximately \$22/oz for a 3-month spread. This carry relationship is the foundation of the gold calendar spread structure.
- **Backwardation in gold:** genuine backwardation (forward below spot) is extremely rare in gold and is always a signal of acute physical market stress — specifically when gold is in extreme demand for immediate delivery and available lease rates rise sharply. Brief backwardation occurred during the September 1999 central bank gold sales panic and intraday during March 2020 COVID liquidity event.

3.2 The EFP (Exchange of Futures for Physical) — The Key Spread

The EFP (or "basis") between COMEX GC futures and LBMA spot gold is the most important spread in the gold market. It reflects: the theoretical carry cost between spot and the futures contract, adjusted for supply/demand imbalances between the two markets:

- **Normal EFP:** COMEX futures trade at a premium to LBMA spot equal to approximately $(\text{USD rate} - \text{lease rate}) \times \text{spot} \times \text{days}/360$. This is the "theoretical" EFP. A premium of \$5–15/oz for a 1-month spread is typical in a 4–5% USD rate environment.
- **Wide EFP (COMEX >> LBMA spot):** a large COMEX premium over LBMA spot signals that financial demand is pushing COMEX futures up relative to the physical market. This can occur when large fund inflows into COMEX futures outpace physical demand (speculative buying without physical backing). Arbitrageurs convert physical gold (buy LBMA, sell COMEX) to compress the EFP.
- **Narrow or negative EFP (COVID March 2020):** in March 2020, the COMEX-LBMA spread inverted — COMEX futures briefly traded above LBMA spot by an unusually large margin, then physical demand in London tightened dramatically as markets dislocated. Simultaneously, logistical constraints (COMEX-approved refineries in Switzerland shutting for COVID) prevented physical gold from being refined into COMEX-deliverable bars, widening the EFP to historically extreme levels (\$50–80/oz). This is the most significant EFP dislocation in modern gold market history.

3.3 Gold Calendar Spreads — The Interest Rate Spread

COMEX gold calendar spreads (e.g. December vs June) are fundamentally interest rate spreads dressed in gold clothing. The driver of gold spreads is changes in the USD interest rate level and the gold lease

rate:

- **When USD rates rise:** the contango widens. Deferred contracts rise relative to nearby (the futures curve steepens in absolute dollar terms). Calendar spreads (deferred minus nearby) widen.
- **When USD rates fall:** the contango narrows. Calendar spreads compress. In extreme low-rate environments (2020-2021), spreads became very narrow because the carry cost of holding gold forward was minimal.
- **When lease rates spike:** the contango narrows or disappears even without a change in USD rates, because the "cost" component of carry falls (lease income offsets USD financing costs). Lease rate spikes typically accompany physical gold scarcity events.
- **Practical spread sizes:** a typical 2-month COMEX gold spread (e.g. June-August) in a 5% USD rate, 0.5% lease rate environment is approximately \$16-20/oz. This narrows to approximately \$3-5/oz in a near-zero rate environment. The spread is highly sensitive to rate changes.

Gold calendar spreads are therefore a direct expression of the money market cost of carry. Trading gold spreads is effectively taking a view on the short-term USD interest rate, the gold lease rate, and their interaction.

4. Real Interest Rates and Central Bank Monetary Policy

Monetary policy is the single most important driver of gold prices over the medium term (3 months to 3 years). The Federal Reserve's rate decisions and balance sheet policy directly determine the real interest rate environment that is the primary valuation framework for gold.

4.1 The Federal Reserve and Gold

- **Fed Funds Rate and real rates:** the Fed Funds Rate determines the nominal short-term rate. The real rate = nominal rate minus inflation expectations. When the Fed hikes rates faster than inflation expectations rise, real rates increase — bearish for gold. When the Fed cuts rates or inflation expectations rise faster than nominal rates, real rates fall — bullish for gold.
- **Fed balance sheet (QE/QT):** QE (Quantitative Easing) — Fed purchases of Treasuries and MBS — depresses real yields and is associated with USD weakness, both bullish for gold. QT (Quantitative Tightening) has the opposite effect. The 2020-2021 QE programme was a primary driver of gold's rally to record highs (\$2,075/oz in August 2020).
- **Yield Curve Control (YCC):** when a central bank caps long-term yields (as the Fed did in WWII and as the Bank of Japan implemented from 2016), real rates are pinned below their natural level while inflation can run higher — a highly gold-bullish environment.
- **The "pivot" trade:** gold often rallies when markets anticipate a shift from Fed tightening to easing (the "pivot"), even before actual rate cuts materialise. This forward-pricing of the pivot makes FOMC communications and economic data that shift pivot expectations the most important near-term gold price drivers.

4.2 TIPS — The Real Rate Signal

- **US Treasury Inflation-Protected Securities (TIPS):** the primary market instrument for measuring real interest rates. The 10-year TIPS yield is the most closely tracked real rate metric for gold pricing.
- **Real yield range and gold price historical relationship:**

10-Year TIPS Yield Range	Historical Gold Price Tendency	Notes
Above +2.0%	Structurally bearish. Gold tends to trade at or below \$1,200–1,500/oz	High real rates make holding gold expensive. 2022 peak real yields (\$1,620/oz trough).
+1.0% to +2.0%	Neutral to modestly bearish. Price range varies.	Normal pre-GFC environment. Gold at long-run cost of production support.
0% to +1.0%	Neutral. Gold in equilibrium range.	Mild positive real rates do not strongly incentivise or penalise holding.
-0.5% to 0%	Mildly bullish. Gold outperforms vs real assets.	Post-2016 environment; gold rallied to \$1,500–1,800/oz.
Below -0.5%	Strongly bullish. Opportunity cost of gold is negative.	2020–2021: real yields at -1.0%; gold peaked at \$2,075/oz. Holding gold costs less than holding TIPS (which yield below zero in real terms).

4.3 Other Major Central Banks

- **European Central Bank (ECB):** ECB policy affects EUR/USD, which is the largest component of the DXY. When the ECB is more dovish than the Fed (cutting rates or expanding balance sheet while the Fed is on hold), EUR weakens and DXY rises — bearish for gold. ECB deposit rate below zero (as occurred 2014–2022) reduces the opportunity cost of gold for Euro-based investors.

- **Bank of Japan (BoJ):** BoJ policy has been the most extreme globally. Years of zero/negative interest rates and YCC created conditions where gold in JPY terms outperformed dramatically. Japanese retail investors (who can buy JPY-denominated gold) have been significant physical gold buyers. BoJ rate normalisation (2024) is a structural shift with implications for gold in JPY terms and for global carry trade unwinds that can affect gold.
- **Global rate coordination:** when central banks globally move in the same direction (e.g. simultaneous post-COVID hiking cycles 2022–2023), the real rate impact on gold is amplified. Global real rate convergence is more important than any single central bank.

5. Central Bank Gold — The Structural Demand Pillar

Central bank demand for gold has been one of the most important structural developments in the gold market over the past 15 years. After decades of being net sellers of gold (1990s–2009), central banks globally have been persistent net buyers since 2010, and this shift in official sector behaviour has had a lasting impact on the gold price structure.

5.1 Official Sector Gold Holdings

- **Total official sector gold holdings:** approximately 35,000–37,000 tonnes globally (approximately 17–18% of total above-ground gold stock). The US Federal Reserve holds the largest single reserve (approximately 8,133 tonnes at Fort Knox, West Point, and the NY Federal Reserve). Other large holders: Germany (~3,352 t), IMF (~2,814 t), Italy (~2,452 t), France (~2,437 t), Russia (~2,330 t), China (~2,200 t reported, likely higher given known PBoC reserve diversification).
- **Net purchases since 2010:** emerging market central banks — particularly from China, Russia, Turkey, India, Poland, Hungary, Kazakhstan, and others — have been the primary buyers. Annual net central bank purchases reached a record ~1,136 tonnes in 2022 and ~1,037 tonnes in 2023 (World Gold Council data). This structural demand has absorbed a significant portion of annual mine supply and supported prices.
- **Motivations for central bank buying:** (1) diversification away from USD-denominated reserves (de-dollarisation, particularly post-2022 Russian asset freezes); (2) gold as a safe haven asset with no counterparty risk; (3) gold as collateral for international settlements; (4) historical monetary role and domestic political considerations in some countries.

5.2 The Washington Agreement and Its Legacy

- The Central Bank Gold Agreement (CBGA, first signed September 1999) was a series of agreements among European central banks to limit the quantity of gold they collectively sold each year (initial cap: 400 tonnes/year). It was signed after large IMF and UK gold sales (most controversially, the UK sold approximately 400 tonnes in 1999–2002 near historical lows — a decision that became known as "Brown's Bottom"). The CBGA was not renewed after its 5th iteration because European central banks had ceased being net sellers.
- The transition from Western central banks as net sellers to EM central banks as net buyers represents a structural shift in the gold supply/demand balance that persists today.

5.3 Central Bank Purchases as a Price Signal

- Large disclosed central bank purchases (particularly from China, Poland, Turkey, India) are market-moving events. China's PBoC reports gold reserve changes monthly (though disclosures are irregular and actual holdings may exceed reported figures). When the PBoC resumes buying after a pause, gold rallies on the disclosure.
- The World Gold Council (WGC) compiles quarterly central bank buying data. The Q4 WGC Central Banks Gold Demand report (published January–February of the following year) provides the annual buying total and is market-relevant.

6. Physical Gold Demand — Jewellery, Investment, and Industry

Physical gold demand encompasses jewellery fabrication, bar and coin investment demand, and industrial/technology uses. While these physical demand components are significant, they are less important as price drivers in the short to medium term than the macro-financial and monetary policy variables described in previous sections. However, they define the structural demand floor and explain why gold prices have not fallen to pure financial asset valuation levels.

Demand Category	Annual Volume (approx.)	Key Markets	Price Sensitivity	GC Curve Impact
Jewellery fabrication	~1,900–2,200 t/yr	India (~25%), China (~25%), Middle East (~10%), other Asia/global	Price-elastic at high prices (demand falls sharply above ~\$2,000/oz historically); income-elastic in EM	Provides a demand floor at lower price levels. Weak at high prices — a bearish signal at extremes.
Bar and coin (retail investment)	~900–1,200 t/yr	China, India, Europe (Germany large bar demand), US, Middle East	Counter-cyclical to some extent — rises in periods of financial uncertainty; falls when confidence high	High bar/coin demand during stress events supports physical floor. ETF demand is a separate, larger signal.
Gold ETFs (investment)	Net flow varies widely (–400 t to +900 t/yr)	SPDR Gold Trust (GLD, NYSE), iShares Gold Trust (IAU), and others globally	Largest and most price-sensitive investment demand category. ETF flows are a real-time signal.	Weekly ETF holding changes (published by ETF providers) are a primary demand indicator for GC mid-curve.
Central bank (official sector)	~1,000–1,100 t/yr (recent record years)	Emerging market central banks (China, Russia, India, Turkey, Poland, Kazakhstan, others)	Less price-elastic — strategic purchases. Can accelerate in response to USD weaponisation risk.	The most structurally important demand category for price support. Quarterly WGC data is the primary source.
Technology/industrial	~300–340 t/yr	Electronics (bonding wire, connectors), dentistry, catalysts, space applications	Relatively price-inelastic (small quantities, high-value applications). No substitution at these uses.	Stable baseline. Not a price driver at the macro level but confirms gold's industrial non-substitutability.

6.1 Gold ETF Flows — The Most Timely Physical Demand Signal

- **SPDR Gold Shares (GLD):** the world's largest gold-backed ETF, managed by State Street with gold custodied by HSBC London. GLD holdings are published daily and represent a real-time measure of institutional and retail investment demand from Western financial markets. GLD AUM peaked at approximately 1,350 tonnes in 2020. Large outflows (100+ tonnes over a few weeks) reflect investor liquidation of gold exposure and are a bearish signal; large inflows are bullish.
- **iShares Gold Trust (IAU):** the second-largest US gold ETF. Also publishes daily holdings. Combined GLD+IAU flows are tracked by WGC and Bloomberg.
- **European gold ETFs:** Xetra Gold (Germany), WisdomTree Physical Gold (UK/Europe), and others provide European retail and institutional demand. German retail demand for physical-backed ETC (Exchange-Traded Commodities) is particularly significant — Germany has a strong culture of gold ownership as savings protection.
- **Chinese domestic gold funds:** Shanghai Gold Exchange (SGE) products and domestic Chinese gold funds add a China-specific investment demand layer that is less visible to Western market participants.

6.2 Indian and Chinese Jewellery Demand — The Price Elasticity

- India and China together account for approximately 50% of global jewellery demand. Both markets are highly price-sensitive:
- **India:** gold jewellery demand is strongly correlated with agricultural incomes (monsoon outcome), the INR/USD rate (high INR/USD = expensive gold in rupees = demand falls), and the wedding season (October–November and April–May peaks). Import duties on gold in India (currently 15% total including GST) create a significant wedge between international prices and domestic Indian retail prices. Duty changes are policy events that affect Indian demand and therefore global physical flows.
- **China:** Chinese gold demand is correlated with GDP growth and the property market cycle. During periods of property sector stress (2021-2024), Chinese retail investors have shifted savings from property to gold — a structural demand support. Chinese New Year is the peak jewellery and bar demand season.

7. Mine Supply — The Long-Run Cost Anchor

Annual gold mine production (approximately 3,600–3,800 tonnes/year) represents roughly 1.5–2.0% of the total above-ground stock. This small flow relative to stock means that mine supply changes have a limited short-to-medium-term price impact compared to shifts in investment demand. However, mine supply economics define the long-run cost of production anchor for prices.

7.1 Major Gold Producing Countries

Country	Annual Production	Key Mines/Regions	Cost Structure	GC Relevance
China	~370 t/yr	Shandong, Inner Mongolia, Xinjiang	High all-in sustaining cost (AISC) ~\$1,400–1,600/oz; limited exports — most consumed domestically	Largest producer but minimal export market relevance. Chinese domestic mine production stays in China.
Russia	~330 t/yr	Siberia, Urals, Far East	AISC ~\$900–1,200/oz; competitive cost structure. Post-2022 sanctions reduced Western market access.	Russian gold increasingly sold to China, Middle East, UAE. Limited COMEX/LBMA delivery. Sanctions created market segmentation.
Australia	~320 t/yr	Western Australia (Super Pit, Boddington), Northern Territory	AISC ~\$1,100–1,400/oz; world-class mines; LBMA good delivery bars produced	Major LBMA-deliverable supply source. Australian production trends affect global physical supply.
Canada	~195 t/yr	Ontario, Quebec, British Columbia	AISC ~\$1,100–1,400/oz; mature mining jurisdiction	LBMA/COMEX deliverable. Major publicly listed producers (Barrick, Agnico Eagle) are Canadian.
USA	~170 t/yr	Nevada (Carlin Trend, Cortez, Goldstrike)	AISC ~\$1,100–1,300/oz; Nevada dominant	COMEX-deliverable. Nevada mines include some of the lowest-cost deposits globally.
Ghana, South Africa, Tanzania, others (Sub-Saharan Africa)	~200 t combined	Various	AISC variable; labour and energy costs significant; sovereign risk	LBMA-deliverable; relevant for global supply balance.
Peru, Mexico, Indonesia (other major)	~300 t combined	Various	Variable; Peru (Yanacocha) and Mexico (Penasquito) among largest	Contribute to global supply balance; less directly relevant to COMEX spot.

7.2 All-In Sustaining Cost (AISC) — The Price Floor

- **AISC definition:** the World Gold Council introduced the AISC metric in 2013 as the comprehensive measure of the cost of producing and sustaining gold production. It includes: mining operating costs, sustaining capex, G&A, royalties, reclamation costs, and by-product credits. It does not include growth capex or exploration.
- **Industry-wide AISC:** approximately \$1,200–1,400/oz for the global industry average (weighted by production). The 90th percentile cost (the most expensive producers) is approximately \$1,600–1,800/oz. These cost levels provide a structural floor for gold prices: sustained prices below industry AISC lead to mine curtailments, deferrals, and closures, reducing supply and supporting prices over 1–3 years.
- **Gold mine production response to price:** unlike oil shale, gold mine production responds very slowly to price signals. Developing a new gold mine takes 7–15 years from discovery to first production. This long lead time means that price spikes do not quickly incentivise new supply —

reinforcing gold's monetary asset characteristics.

- **Mine hedging (producer hedge books):** gold mining companies have historically hedged a portion of forward production by selling gold forwards (short COMEX or OTC forward contracts). The total industry "hedge book" (the net short forward position) shrank dramatically from approximately 3,000 tonnes in 2000 to approximately 100–200 tonnes today, as low prices in the early 2000s and then rising prices made hedging less attractive. Producer de-hedging (buying back forward shorts) was a significant source of demand from 2000–2010. Current hedge book size is tracked by WGC and Metals Focus.

7.3 Secondary Supply — Recycling

- **Gold recycling:** approximately 1,100–1,300 tonnes/year of gold is recycled globally, primarily from jewellery scrap (90%+) and a small amount from electronics. Recycling is the second-largest source of gold supply after mine production.
- Recycling volumes increase at high prices (consumers sell old jewellery when the gold price is elevated), providing a mild supply-side negative feedback. Conversely, at low prices, recycling falls. This self-correcting mechanism reduces price volatility at extremes.

8. Geopolitical Risk and Safe Haven Demand

Gold's role as a safe haven asset — a store of value that maintains purchasing power through financial crises, wars, and political instability — is one of the most empirically documented features of the market. Geopolitical events and financial stress generate demand that is separate from and can override the real rate framework in the short term.

8.1 Documented Safe Haven Episodes

Event	Period	Gold Price Response	Duration of Premium	Key Mechanism
Gulf War I	Aug 1990 – Jan 1991	Rally ~\$50-80/oz then faded	Weeks	Geopolitical supply risk premium; faded as war was brief
9/11 attacks	Sep 2001	Initial spike ~\$10/oz; modest	Days	Safe haven; limited persistence as financial system held
GFC / Lehman	Sep 2008 – Feb 2009	Initial fall (liquidity), then strong rally	Years (2008-2011)	Currency debasement fears; QE expectations; USD collapse fears
European Debt Crisis	2010-2012	Gold to \$1,920 (then record)	2 years	EUR existential risk; safe haven vs fiat currency system
Ukraine/Russia I	Mar 2014	+\$50-60/oz spike, then partially faded	Weeks-months	Geopolitical tension in Europe; Russia gold buying increased
COVID-19 pandemic	Mar-Aug 2020	Initial fall, then record \$2,075/oz	1 year	QE expectations; negative real rates; currency debasement fears
Russia-Ukraine II	Feb 2022	+\$100-150/oz initial; sustained elevated	Months-ongoing	Sanctions on Russian FX reserves → CB de-dollarisation acceleration
Middle East escalation (2023-)	Oct 2023-	Significant rally, sustained above \$2,000/oz	Ongoing	Regional war risk; USD alternative demand; CB buying continues

8.2 The De-Dollarisation Theme

- The February 2022 freezing of approximately \$300 billion in Russian central bank foreign exchange reserves (held primarily in USD, EUR, and GBP) was a watershed event for the global monetary system. It demonstrated that USD-denominated reserve assets are subject to confiscation risk from their issuing sovereigns in the context of geopolitical conflict.
- This event accelerated the already-ongoing trend of EM central banks diversifying reserves into gold — which cannot be frozen or confiscated once it is physically held within a country's own vaults. The 2022 and 2023 record central bank gold purchases (1,136 and 1,037 tonnes respectively) directly reflect this de-dollarisation motivation.
- The de-dollarisation trend is a structural multi-year driver that has permanently shifted the central bank gold demand outlook. This is the most important long-run fundamental change in the gold market of the past decade.

8.3 The VIX and Financial Stress Indicators

- **VIX (CBOE Volatility Index):** gold and the VIX have a positive but non-linear relationship. Moderate VIX spikes (20-30) are associated with gold holding steady or rising mildly. Extreme VIX spikes (40+, March 2020) can initially cause gold to fall as investors sell liquid assets (including gold) to meet margin calls — the "liquidity crunch" dynamic. But gold typically recovers quickly as the crisis triggers policy responses (rate cuts, QE) that are gold-bullish.

- **Credit spreads:** widening investment-grade and high-yield credit spreads signal financial stress. The gold-credit spread relationship is similar to VIX — moderate stress is gold-bullish; extreme stress initially may cause gold to sell as a liquid asset, then recover.
- **Financial Stress Index (Chicago Fed NFI):** when the NFI moves above zero (financial tightening), gold tends to benefit from safe-haven flows. Extreme tightening above +1 can create the liquidity-sell dynamic briefly.

9. Cross-Market Relationships

9.1 Gold and the US Dollar (DXY) — The Primary Cross-Market Signal

- **The negative correlation:** empirically, the daily correlation between DXY and gold is approximately -0.5 to -0.7 over most market periods. The mechanism: gold is priced in USD; a 1% rise in DXY makes gold approximately 1% more expensive in all non-USD currencies, reducing demand. The correlation is tighter when DXY moves are driven by US real rate differentials vs other countries.
- **The correlation breaks at extremes:** in risk-off episodes where both USD and gold are sought as safe havens simultaneously, or in inflationary episodes where USD is inflated and gold is held as a hedge, the two can move in the same direction. This break in normal correlation is itself a signal — it indicates a regime shift in investor priorities.

9.2 Gold and TIPS Yields — The Primary Fundamental Spread

- The spread between gold prices and the level implied by a regression model of gold vs 10-year TIPS yields is the primary valuation gap metric used by macro hedge funds and gold analysts. When gold trades significantly above the TIPS-implied level, it suggests either: (1) the safe-haven/geopolitical premium is elevated; (2) markets expect real rates to fall further; or (3) there is a structural demand shift (central bank buying decoupling gold from rate sensitivity).
- The growing central bank demand since 2022 has measurably shifted the gold-TIPS relationship — gold has traded at a larger premium to TIPS-implied levels than in previous cycles, suggesting that the structural central bank demand has raised the equilibrium price range.

9.3 Gold and Silver — The Gold-Silver Ratio

- **The Gold-Silver ratio (GSR):** gold price divided by silver price. The ratio has historically ranged from approximately 40:1 to 120:1 in modern markets, with a long-run average near 60-70:1. When the GSR is extreme (above 90-100:1), silver is historically cheap relative to gold; when below 50:1, silver is expensive relative to gold. The GSR is used as a mean-reversion tactical signal by precious metals spread traders.
- In recessions and financial stress, the GSR tends to spike (gold outperforms as the purer safe haven; silver falls due to its larger industrial demand component). In economic recoveries with inflationary impulse, silver tends to outperform gold (higher beta industrial-precious hybrid). The GSR therefore also encodes a macro risk appetite signal.

9.4 Gold and US Equities (S&P; 500)

- The gold-S&P; 500 relationship is regime-dependent: in normal economic expansions, equities outperform gold and the ratio (Dow-to-Gold or S&P-to-Gold;) rises. In recessions and financial crises, gold outperforms equities as a safe haven. The ratio of gold to equities is a long-run mean-reverting relationship that provides a very long-run valuation context (though not a tactical trading signal).
- Gold and equities can be positively correlated in "risk-on" periods where both are rising on dollar weakness and easy money expectations. They are negatively correlated in crisis periods where gold is a safe haven against falling equities.

9.5 Gold and Crude Oil

- The gold-oil ratio (gold price / WTI price) historically ranges from approximately 10:1 to 30:1. High gold-oil ratio = gold expensive relative to oil (often in recessions when oil falls sharply). Low gold-oil ratio = gold cheap relative to oil (often in supply-constrained inflationary energy cycles). The ratio is a macro regime indicator rather than a direct price signal.

- The direct oil-gold price relationship is indirect: higher oil prices can increase inflation expectations, which can support gold if they lead to lower real rates. But the mechanism requires a policy response that does not fully offset inflation via rate hikes.

10. Forward Curve Zones and Spread Structure

The COMEX gold forward curve is in permanent contango (in normal conditions) because gold has positive carry (cost of financing exceeds lease income). The curve's shape and the absolute level of spreads are primarily determined by the short-term USD interest rate. This makes the gold curve unique among commodity forward curves — it is driven by monetary policy, not seasonal supply/demand.

Zone	Contracts	Primary Driver	Secondary Driver	Key Signal	Volatility
Ultra-front (spot-to-first)	GC first active vs LBMA spot	EFP (physical vs futures basis); short-term lease rates; COMEX inventory and delivery queue	Physical gold flows between New York and London; COMEX warehouse stock levels	Daily EFP level; COMEX eligible and registered warehouse stocks	Low normally; extreme in stress events
Front (C1–C4)	~0–12 months forward	Fed Funds rate / SOFR; near-term monetary policy expectations; TIPS yields (2-year most relevant)	Gold lease rates; FOMC meeting outcomes; USD (DXY); ETF flow direction; safe haven demand spikes	2-year TIPS yield; FOMC decisions; weekly ETF flows; spot-equivalent DXY level	High
Belly (C4–C8)	~12–24 months forward	Expected monetary policy path (dot plot equivalent for gold); 5–10 year real rate expectations; USD secular trend	Central bank gold demand (quarterly WGC data); gold mine supply outlook; inflation expectations	10-year TIPS yield; SEP dot plot rate path; WGC quarterly demand report	Moderate
Back curve (C8+)	24+ months forward	Long-run neutral real rate expectations; structural inflation outlook; de-dollarisation and reserve diversification trends	Mine supply capex cycle; long-run demand growth (EM middle class, CB de-dollarisation); energy transition gold demand	IEA/IMF long-run rate scenarios; WGC long-term demand research; major central bank reserve policy signals	Low

10.1 The Gold Butterfly

- A gold butterfly (long C1 + long C3 – 2×C2) on the COMEX curve captures the curvature of the carry structure at a specific point. In normal markets, the gold curve is very nearly linear (carry adds at a constant rate per month), so butterflies are near zero.
- Butterflies become non-zero and tradeable when: (1) the yield curve has a non-linear shape (e.g. inversion at a specific tenor), because gold carry follows the shape of the USD rate curve; (2) a specific FOMC decision is expected within the belly contract but not the wings; (3) a gold-specific lease rate spike creates a non-linear effect at a specific point in the forward curve.
- Gold butterflies are therefore functionally similar to SOFR butterflies — they price the curvature of the interest rate path at a specific horizon. Understanding the FOMC meeting calendar relative to the contract reference periods is the foundation of gold butterfly positioning.

10.2 Seasonal Patterns in Gold

- **Physical demand seasonality:** there is a weak but documented seasonal pattern in physical gold demand. Indian wedding seasons (October–November, April–May) and Chinese New Year (January–February) create modest seasonal demand increases for jewellery. However, these seasonal patterns are overwhelmed by macro and financial drivers in most years and are not reliable tactical signals.
- **Year-end positioning:** gold sometimes benefits from year-end portfolio rebalancing (investors adding gold as a portfolio hedge). The effect is statistically weak and inconsistent.

- **The absence of strong seasonality** is itself a characteristic that distinguishes gold from agricultural and energy markets. Do not apply seasonal pattern analysis to gold with the same confidence as for natural gas or RBOB.

11. COMEX Positioning, Microstructure, and Data

11.1 Commitments of Traders (COT)

- **Managed money net position:** the most widely watched positioning metric for gold. Managed money (hedge funds, CTAs, macro funds) are the primary speculative participants in COMEX gold. Extreme net long positioning (above 200,000–250,000 contracts) signals crowding and elevated reversal risk on any bearish data surprise. Extreme net short positioning (below –30,000 to –50,000 contracts, rare but occurred in 2018 and late 2022) signals a short-covering rally potential on any positive catalyst.
- **Commercial positioning (swap dealers and producers):** commercial shorts represent mine producer hedging and bank market-making positions. A large commercial short position signals that producers have locked in prices — a reflection of producer confidence in elevated prices but also a source of known future supply.
- **Non-commercial long concentration:** when a small number of large non-commercial traders hold a concentrated long position, the market is susceptible to rapid liquidation if those holders decide to exit simultaneously.

11.2 COMEX Warehouse Stocks

- **Registered gold:** gold in COMEX-approved warehouses that has been "registered" — i.e. certified for delivery against a futures contract. A decline in registered stocks signals that deliverable supply is being depleted — potentially bullish for the front contract if open interest is high relative to registered stocks.
- **Eligible gold:** gold in COMEX-approved warehouses that meets delivery standards but has not been registered. Can be converted to registered by the holder. Total warehouse stocks = registered + eligible.
- **Warehouse stock movements:** large inflows or outflows from COMEX warehouses (published daily by CME) reflect arbitrage activity between COMEX and LBMA markets. March 2020 saw a dramatic rush to move LBMA-format bars to COMEX-deliverable bars as the EFP widened and futures buyers demanded physical delivery.

12. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Release Timing	Primary GC Curve Impact
US CPI (headline + core)	BLS	Monthly	~2nd week, 08:30 ET	All contracts — most important monthly macro event for gold; drives real rate expectations
US Core PCE Price Index	BEA	Monthly	~4th week, 08:30 ET	C2-C8 — Fed's primary inflation target; PCE surprise adjusts real rate and gold valuation
FOMC rate decision + statement	Federal Reserve	8× per year	2:00 PM ET	All contracts — immediate repricing of carry structure and gold valuation framework
FOMC Chair press conference	Federal Reserve	8× per year	2:30 PM ET	All contracts — often moves gold more than the statement itself
SEP / Dot Plot	Federal Reserve	4× per year (Mar/Jun/Sep/Dec)	2:00 PM with FOMC	C4-C12 — long-run rate path shifts the carry structure and gold valuation anchor
10-year TIPS yield (continuous)	Bloomberg / Reuters / Fed	Continuous	Market hours	All contracts — the single most important daily input to gold pricing model
US Dollar Index (DXY)	ICE / Bloomberg	Continuous	Market hours	All contracts — negative correlation; USD move is the most correlated single variable
Non-Farm Payrolls + Unemployment	BLS	Monthly	1st Friday, 08:30 ET	C1-C4 — strong jobs data → higher real rates → gold bearish; weak data → gold bullish
LBMA Gold Price AM/PM Fix	IBA via LBMA	Daily	10:30 / 15:00 London	Spot reference — PM fix most used for ETF NAV and physical transactions
COMEX EFP (Exchange of Futures for Physical)	CME / bullion banks	Continuous	Market hours	Front contract — key signal of physical vs paper gold dislocation; stress indicator
COMEX warehouse stocks (registered + eligible)	CME Group	Daily	After close	Front contract — registered stocks vs open interest determines delivery squeeze risk
Gold ETF daily holdings (GLD, IAU, others)	State Street, BlackRock, others	Daily	After close / morning next day	C1-C4 — investment demand signal; large weekly outflows/inflows are primary demand events
WGC Quarterly Gold Demand Trends	World Gold Council	Quarterly	~4-6 wks after quarter end	C4-C12 — central bank buying data; jewellery, bar/coin, technology breakdown
WGC Central Bank Statistics	World Gold Council	Monthly	~3rd week of month	C2-C8 — monthly central bank purchase disclosures; PBoC report is most watched
Jackson Hole speech (Fed Chair)	Federal Reserve	Annual (August)	~10:00 AM ET	All contracts — major policy pivot signals have originated here; gold highly reactive
CFTC COT (managed money, commercial)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — extreme positions as contrary indicator

IMF / WGC official gold holdings data	IMF / WGC	Monthly	~Mid-month	C4-C12 — structural central bank reserve data; PBoC monthly disclosure most watched
ICAC / Metals Focus mine supply data	Metals Focus / GFMS / WGC	Annual + quarterly	Variable	C8+ — long-run supply anchor; AISC trend affects gold long-run floor pricing

COMEX Gold is the most macro-financial of all the markets covered in this document series. Unlike petroleum, agriculture, or base metals, gold's price is not anchored by consumption cycles, physical production constraints, or seasonal demand patterns in any meaningful short-to-medium-term sense. It is anchored by the real interest rate — the opportunity cost of holding a non-yielding monetary asset — and by the confidence of investors and central banks in the global monetary and financial system.

The forward curve and calendar spreads in gold are functionally interest rate instruments: the contango between contracts equals the net carry cost (USD short rate minus gold lease rate). Trading gold calendar spreads is therefore a direct position in the near-term USD interest rate relative to the gold lease rate — a short-term rates product, not a commodity product. Butterflies on the gold curve reflect the same FOMC meeting calendar dynamics as SOFR butterflies. The edge in gold spread trading comes from correctly anticipating changes in the short-term USD rate, the gold lease rate, and the occasional physical dislocation (EFP stress events) that creates non-carry-based spread opportunities.